

Indian Institute of Technology Kanpur
MTH308 & MTH308B: Numerical Analysis and Scientific
Computing-I
January-April 2024

Quiz-1

Full marks: 40

Date: 03.02.2024

Duration: 1 hours 30 mins

If you are using any result or theorem, please write the statement of that result/theorem clearly.

Answer all questions.

$$(2 + 2 + 2) + 3 + 3 + (3 + 3) + 2 + 5 + (5 + 2) + (4 + 4) = 40$$

1. Justify your answer (no marks will be given without proper justification):
 - (i) True or false: A good algorithm produces an accurate solution regardless of the condition of the problem being solved.
 - (ii) Floating-point addition is associative but not commutative.
 - (iii) For a nonsingular matrix $A \in M_n(\mathbb{R})$, $\mathcal{K}(cA) = c\mathcal{K}(A)$, for all $c \in \mathbb{R} \setminus \{0\}$. Here $\mathcal{K}(\cdot)$ denotes the condition number of a matrix.

2. Give an example to show that the floating-point multiplication is not necessarily associative.

3. Let $x = 2$, $y = -0.6$ and $z = 0.602$. Using 3-digit rounding compute

$$(x \times y) + (x \times z) \quad \text{and} \quad x \times (y + z).$$

What is your conclusion about the property of floating point arithmetic in the above computation?

4. Find the number of significant digits in x_A for approximating x of the following.

(i) $x = -0.04518$ $x_A = -0.045113$.

(ii) $x = 23.4604$ $x_A = 23.4213$.

5. Find the condition number at a point $x = c$ for the function $f(x) = \pi^x$.

6. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two continuously differentiable functions such that

- there exists constant $M > 0$ such that

$$|f'(x)| \geq M \quad \text{and} \quad |g'(x)| \leq M \quad \text{for all } x \in \mathbb{R},$$

- the process of evaluating f is well-conditioned,

- the process of evaluating g is ill-conditioned.

Show that $|g(x)| < |f(x)|$ for all $x \in \mathbb{R}$.

7. Find LU factorization of the given matrix using Gaussian elimination

$$A = \begin{bmatrix} 1 & 1 & -1 & 2 \\ 1 & -1 & 1 & 7 \\ 2 & 2 & 4 & 5 \\ 2 & 3 & 4 & 6 \end{bmatrix}.$$

Find $\|A\|_1$ and $\|A\|_\infty$ subordinate to the vector norm $\|\cdot\|_1$ and $\|\cdot\|_\infty$ respectively.

8. Find LU factorization of the tri-diagonal matrix

$$B = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix}.$$

Hence, find its Cholesky's decomposition.

Do not write anything on the question paper.

End.

Indian Institute of Technology Kanpur
MTH308 & MTH308B: Numerical Analysis and Scientific
Computing-I
January-April 2024
Midsem

Full marks: 60

Date: 22.02.2024

Duration: 2 hours

If you are using any result or theorem, please write the statement of that result/theorem clearly.

Answer all questions. $(4 + 4 + 5) + (5 + 5 + 4) + (5 + 4 + 10) + (5 + 5 + 4) = 60$

Notation: $n \in \mathbb{N}$. $M_n(\mathbb{R})$ = Set of all $n \times n$ matrices over $\mathbb{R}$.

1. In a floating point number system $\mathbb{F}$, let β (base), n (precision), L (lower bound of the exponent range), U (upper bound of the exponent range) be given with usual meaning.

i) Find (with explanation) UFL and OFL of $\mathbb{F}$.

ii) Let $x \in \mathbb{R}$, $x \neq 0$ be such that $\text{UFL} \leq |fl(x)| \leq \text{OFL}$. Show that

$$E_{ar}(fl(x)) \leq \beta^{-n+1}, \quad \text{in case of chopping.}$$

iii) Let $x, y \in \mathbb{R}$ with $x < 0 < y$. Let x_A and y_A be approximations of x and y such that they have seven and nine significant digits, respectively. Show that $z_A = x_A - y_A$ has at least six significant digits (i.e. to show $r_{z, z_A} \geq 6$) when compared to $z = x - y$.

2. i) Factor the matrix

$$A_1 = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 6 & 2 \\ 2 & 2 & 5 \end{bmatrix},$$

so that $A_1 = LL^T$, where $L \in M_3(\mathbb{R})$ is a lower triangular matrix.

ii) For $Q \in M_n(\mathbb{R})$ non-singular, let $\mathcal{K}_p(Q)$ denote the condition number of Q with respect to the matrix norm subordinate to the vector norm $\|\cdot\|_p$ of $\mathbb{R}^n$ for $1 \leq p \leq \infty$.

a) Let $Q \in M_n(\mathbb{R})$ be such that $Q^T Q = I$, where $I \in M_n(\mathbb{R})$ is the identity matrix. Show that $\mathcal{K}_2(Q) = 1$.

b) Find all diagonal matrices $D \in M_n(\mathbb{R})$ such that $\mathcal{K}_\infty(D) = 1$.

iii) Let $P \in M_n(\mathbb{R})$ be non-singular and $B \in M_n(\mathbb{R})$ satisfy

$$\frac{1}{\|P - B\|} > \|P^{-1}\|,$$

where $\|\cdot\|$ is a matrix norm subordinate to a vector norm in $\mathbb{R}^n$. Show that B is non-singular.

$$\frac{1}{\|P^{-1}\|} = \frac{1}{\|P^{-1}\|} \leq \frac{\|P - B\|}{\|P\|}$$

$$\frac{\|P - B\|}{\|P\|} = \frac{\|P - B\|}{\|P\|}$$

3. Let $A = (a_{ij}) \in M_n(\mathbb{R})$ with $a_{ii} \neq 0$, for $i = 1, 2, \dots, n$, and $v \in \mathbb{R}^n$. Let us consider the system of linear equations

$$Ax = v.$$

- i) Write the SOR iteration (with Gauss-Seidel) in fixed point formulation of the system above by mentioning M_{SOR} and y_{SOR} properly.
 - ii) If SOR iteration converges for any initial vector, show that the parameter, involved in the SOR, lies in the interval $(0, 2)$.
 - iii) For a strictly diagonal dominant matrix A , show that Gauss-Seidel method converges faster than the Gauss-Jacobi method. $\lambda \leq \mu < 1$
4. i) Show that a strictly diagonal dominant $n \times n$ matrix is non-singular. $\checkmark$
- ii) Let $E, F \in M_n(\mathbb{R})$ be strictly diagonal dominant. Which of the following must be strictly diagonal dominant? Justify your answer. (No marks without justification.)

(a) $-E$, (b) E^T , (c) $E + F$, (d) E^2 , (e) $E - F$.

- iii) Prove that every eigen value of the matrix

$$\begin{bmatrix} 6 & 2 & 1 & 1 \\ 1 & -5 & 0 & 2 \\ 2 & 1 & 4 & 0 \\ 1 & -1 & -2 & -6 \end{bmatrix}$$

lies in the annular subset $\{z \in \mathbb{C} : 1 \leq |z| \leq 10\}$ of the complex plane.

$\lambda \in \mathbb{R}, |\lambda| \leq 10$

Do not write anything on the question paper.

End.

Indian Institute of Technology Kanpur
MTH308 & MTH308B: Numerical Analysis and Scientific
Computing-I
January-April 2024

Quiz-2

Full marks: 30

Date: 13.04.2024

Duration: 1 hours 10 mins ,

Answer all questions. $3 + 2 + 2 + (2 + 2 + 2) + (3 + 1) + 4 + 5 + (2 + 2) = 30$

1. Let $\epsilon > 0$ and $f : [1, 3] \rightarrow \mathbb{R}$ be a function such that $f(1)f(3) < 0$. Let $\{x_k\}_{k \in \mathbb{N}}$ be the bisection iterative sequence converging to a zero r of f in $[1, 3]$. How many iterations (k) do you require to get the accuracy $|x_k - r| \leq \epsilon$? (Justify)

2. Let $n \in \mathbb{N}$, $n \geq 2$ and $y > 0$ be given. Write a Newton-Raphson scheme to find $y^{\frac{1}{n}}$.

3. List one advantage and one disadvantage of the secant method compared with the Newton-Raphson method for solving a nonlinear equation of one unknown.

4. Let $r \in \mathbb{R}$ be a root of the equation $f(x) = 0$.

(i) How do you choose a function $g : \mathbb{R} \rightarrow \mathbb{R}$ so that the sequence

$$x_k = g(x_{k-1}), \quad k \in \mathbb{N},$$

is well-defined and locally converges to r .

(ii) What is the rate of convergence? (Justify)

(iii) What additional assumption(s) do you need to get cubic convergence? (Justify)

5. In an IPL match RCB scores the following.

Over	1	2	4	8
Run	3	7	21	73

Using a polynomial interpolation, guess the projected score at over 10. What was the (approximated) score at over 3 ?

6. Consider the following data given in a table below.

x	3	4	1
$f(x)$	-1	2	-2

Using polynomial interpolation, find of $f^{-1}(0)$ (an approximation).

7. Construct a natural cubic spline for the following data :

x	1	2	3
$f(x)$	2	3	5

8. Classify the functions for which the Trapezoidal rule gives accurate result. Do the same for the Simpson $1/3$.

Do not write anything on the question paper.

End.

$$b_i = \frac{1}{h_i} (a_{i+1} - a_i) - \frac{h_i}{3} (c_{i+1} + 2c_i)$$

$$d_i = \frac{1}{3h_i} (c_{i+1} - c_i)$$

$$\frac{3}{h_j} (a_{i+1} - a_i) - \frac{3}{h_{i-1}} (a_i - a_{i-1})$$

Indian Institute of Technology Kanpur
MTH308 & MTH308B: Numerical Analysis and Scientific Computing-I
January-April 2024
Endsem

Full marks: 80

Date: 24.04.2024

Duration: 3 hours

Please write your answer for each question in order, otherwise, it might be missed to evaluate. If you are using any result/theorem, please write the statement of that result/theorem clearly.

Notation: $n \in \mathbb{N}$. $M_n(\mathbb{R})$ = Set of all $n \times n$ matrices over $\mathbb{R}$. $[a, b] \subset \mathbb{R}$ with $a < b$.

1.

3 + 6 + 3 + 3 = 15

- i) Let $x \in \mathbb{R}$, $x \neq 0$. Let $x_A = 2.5$ be an approximation of x with absolute error at most 0.01. The function $f(x) = x^{2024}$ is evaluated at x_A instead of x . Estimate the resulting absolute error.
- ii) Let $B = (b_{ij}) \in M_n(\mathbb{R})$ be a positive definite matrix. Prove that $b_{ij}^2 < b_{ii}b_{jj}$ for all $i, j = 1, 2, \dots, n$ and $i \neq j$.
- iii) Let $A \in M_3(\mathbb{R})$ be given by

$$A = \begin{bmatrix} 4 & \alpha & 1 \\ 2\beta & 5 & 4 \\ \beta & 2 & \alpha \end{bmatrix}$$

Find all values of $\alpha \in \mathbb{R}$ and $\beta > 0$ so that A is strictly diagonal dominant.

- iv) Compute $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ of the Gauss-Seidel iteration for the following system using $\mathbf{x}^{(0)} = (0, 0, 0)^T$.

$$\begin{aligned} 3x_1 - x_2 + x_3 &= 1, \\ 3x_1 + 6x_2 + 2x_3 &= 0, \\ 3x_1 + 3x_2 + 7x_3 &= 4. \end{aligned}$$

$$x_{k+1} = \frac{1}{a_{ii}} \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_{k+1} - \sum_{j=i+1}^n a_{ij} x_k \right)$$

(4 + 3) + 8 = 15

2.

- i) Let $g : [a, b] \rightarrow [a, b]$ be continuously differentiable function satisfying

$$\sup_{x \in [a, b]} |g'(x)| < 1.$$

- Show that for any choice of $x_0 \in [a, b]$, the sequence $\{x_k\}_{k \in \mathbb{N} \cup \{0\}}$ defined by

$$x_k = g(x_{k-1}), \quad k \in \mathbb{N},$$

is a Cauchy sequence.

- Hence, show that $\{x_k\}_{k \in \mathbb{N} \cup \{0\}}$ converges to the unique fixed point r of g in $[a, b]$.
- ii) Let $f : [a, b] \rightarrow \mathbb{R}$ be twice continuously differentiable function satisfying
 - $f(a)f(b) < 0$, $\frac{|f(a)|}{|f'(a)|} < b - a$, and $\frac{|f(b)|}{|f'(b)|} < b - a$
 - for all $x \in [a, b]$, $f'(x) \neq 0$ and $f''(x) \leq 0$

Prove that for any choice of $x_0 \in [a, b]$, the Newton-Raphson sequence converges to the unique solution of the equation $f(x) = 0$ in $[a, b]$.

3. Use Horner's method (synthetic division) to compute $x^{(1)}$ and $x^{(2)}$ of the Newton-Raphson iteration to solve the equation

$$x^5 + 6x^4 - 2x^3 - 10x^2 + 21x - 73 = 0, \quad \text{with } x^{(0)} = -2.$$

6

4. Compute $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ of the Newton iteration to solve the following system of equations

$$x_1^2 + x_2 = 1, \quad x_1 - x_2^2 = 1, \quad x_1 + x_2 + x_3 = 3 \quad \text{with } \mathbf{x}^{(0)} = (0, 0, 0)^T.$$

6

~~7 + 6 + 2 + 1 = 16~~

5.

$$5 + 2 + 5 + 5 = 17$$

- i) Let $\{x_i\}_{i=0}^n \subset [a, b]$ be set of distinct points. Let $f : [a, b] \rightarrow \mathbb{R}$ be C^n . Prove that there exists $\xi \in (a, b)$ such that

$$f[x_0, x_1, \dots, x_n] = \frac{f^n(\xi)}{n!}.$$

- ii) What is Runge phenomenon?

- iii) Let $p(x) := x + 1$, $q(x) := 2x + 1$, and $r(x)$ be interpolating polynomials for the three sets of data

x	0	1
y	y_0	y_1

x	0	2
y	y_0	y_2

and

x	1	2	3
y	y_1	y_2	y_3

respectively. Let $s(x)$ be the interpolating polynomial for the data

x	0	1	2	3
y	y_0	y_1	y_2	y_3

Let $r(2.5) = 10$. Find $s(2.5)$.

- iv) Determine constants $a_i \in \mathbb{R}$ for $i = 0, 1, \dots, 7$ (if exist) so that the function

$$S(x) := \begin{cases} 1 + a_4x, & -4 \leq x \leq -3, \\ a_0 + a_1x + a_2x^2 + a_3x^3, & -3 \leq x \leq 4, \\ a_5 - 32x, & 4 \leq x \leq 5, \end{cases}$$

is a clamped cubic spline interpolating the following data

	0	1	2	3
x	-4	-3	4	5
y	9	7	29	-3
y'	a_6	*	*	a_7

$$\begin{aligned} s'_i(x_0) &= f'(x_0) \\ s'_i(x_n) &= f'(x_n) \end{aligned}$$

6.

5

Let $h > 0$ be fixed. Determine constants $c_i \in \mathbb{R}$ for $i = 1, 2, 3, 4$, (if exist) so that the formula $H(f)$ below

$$\int_0^h f(x) dx \approx H(f) := c_1 f(0) + c_2 f(h) + c_3 f'(0) + c_4 f'(h),$$

produces a quadrature formula that has largest possible degree of precision.

7.

$$3 + 2 + 8 + 3 = 16$$

Let $h > 0$ and $c, c + h, c + 2h \in [a, b]$ and $f : [a, b] \rightarrow \mathbb{R}$ be a C^3 function.

- Find the Divided difference interpolating polynomial approximating f corresponding to the points $c, c + h, c + 2h$.
- Find the (Mathematical) error term in divided difference form for the above approximation. (proof not required)
- Using i) and ii), find the approximation formula for $f'(c)$, the corresponding Mathematical error and Arithmetic error.
- Find the value of h in terms of E, M_j for which absolute Total error is minimized, where E is the machine epsilon and

$$M_j = \sup_{x \in [a, b]} |f^{(j)}(x)|, \quad j = 0, 1, 2, 3.$$

Do not write anything on the question paper.

End.



Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

MTH 636M: Game Theory

Quiz 1, Date: February 06, 2024, Tuesday

Timing: 10:30 AM to 11:50 AM

- Answer any four questions from the following. The exam is for 20 marks.
- Try not to use any result not done in the class. However, if you use any such result, clearly state and prove it.
- Write your name, roll no., program name, and seat number clearly on the top of your answer sheet.
- One A4 sheet with ONLY necessary definitions and results are allowed during the exam. Use of a calculator, mobile, and smart watch is strictly prohibited.

* * * * *

1. Consider the following matrix game:

$$\begin{bmatrix} 3 & 2 & 0 \\ 1 & 2 & 2 \\ 0 & 2 & 4 \\ 0 & 3 & 1 \end{bmatrix}$$

- Determine all the max-min (pure) strategies of both the players. What can you conclude about the value of the game in mixed strategies? (2 marks)
- The value of the game is $\frac{12}{7}$. Use this to give an argument why player 2 will put zero probability on column 2 in any maxmin strategy. (3 marks)

2. Let A and B be two finite-dimensional matrices with positive entries. Show that the game

$$\begin{array}{|c|c|} \hline A & 0 \\ \hline 0 & B \\ \hline \end{array}$$

has no value in pure strategies. (Each 0 here represents a matrix of the proper dimensions, such that all of its entries are 0.) (5 marks)

3. Rohit (Player 1), and Virat (Player 2), are business partners. Each of the partners has to determine the amount of effort he or she will put into the business, which is denoted by e_i , $i = 1, 2$, and may be any non-negative real number. The cost of effort e_i for Player i is ce_i , where $c > 0$ is equal for both players. The success of the business depends on the amount of effort put in by the players; the business's profit is denoted by $r(e_1, e_2) = e_1^{\alpha_1} e_2^{\alpha_2}$, where $\alpha_1, \alpha_2 \in (0, 1)$ are fixed constants known by Rohit and Virat, and the profit is shared equally between the two partners. Each player's utility is given by the difference between the share of the profit received by that player and the cost of the effort he or she put into the business. Answer the following questions:

(i) Describe this situation as a strategic-form game. (2 marks)

(ii) Find all the Nash equilibria (in pure strategies) of the game. (3 marks)

4. Two competing coffee house chains, Pete's Coffee and Caribou Coffee, are seeking locations for new branch stores in Cambridge. The town is comprised of only one street, along which all the residents live. Each of the two chains therefore needs to choose a single point within the interval $[0, 1]$, which represents the exact location of the branch store along the road. It is assumed that each resident will go to the coffee house that is nearest to his place of residence. If the two chains choose the exact same location, they will each attract an equal number of customers. Each chain, of course, seeks to maximize its number of customers.

To simplify the analysis required here, suppose that each point along the interval $[0, 1]$ represents a town resident, and that the fraction of residents who frequent each coffee house is the fraction of points closer to one store than to the other.

(i) Describe this situation as a two-player strategic-form game. (1.5 marks)

(ii) Prove that the only equilibrium in this game is that given by both chains selecting the location $x = \frac{1}{2}$. (1.5 marks)

(iii) Prove that if three chains were to compete for a location in Cambridge, the resulting game would have no equilibrium. (Under this scenario, if two or three of the chains choose the same location, they will split the points closest to them equally between them.) (2 marks)

- ✓ 5. Consider the following two-player zero-sum game:

	L	M	R
A	2	3	$\frac{7}{3}$
B	3	1	$\frac{7}{3}$

~~(i)~~ Find the value of the game and a max-min strategy for both the players. (3 marks)

~~(ii)~~ Describe all the max-min strategies for Player 2 (Column player). (2 marks)



Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

MTH 636M: Game Theory

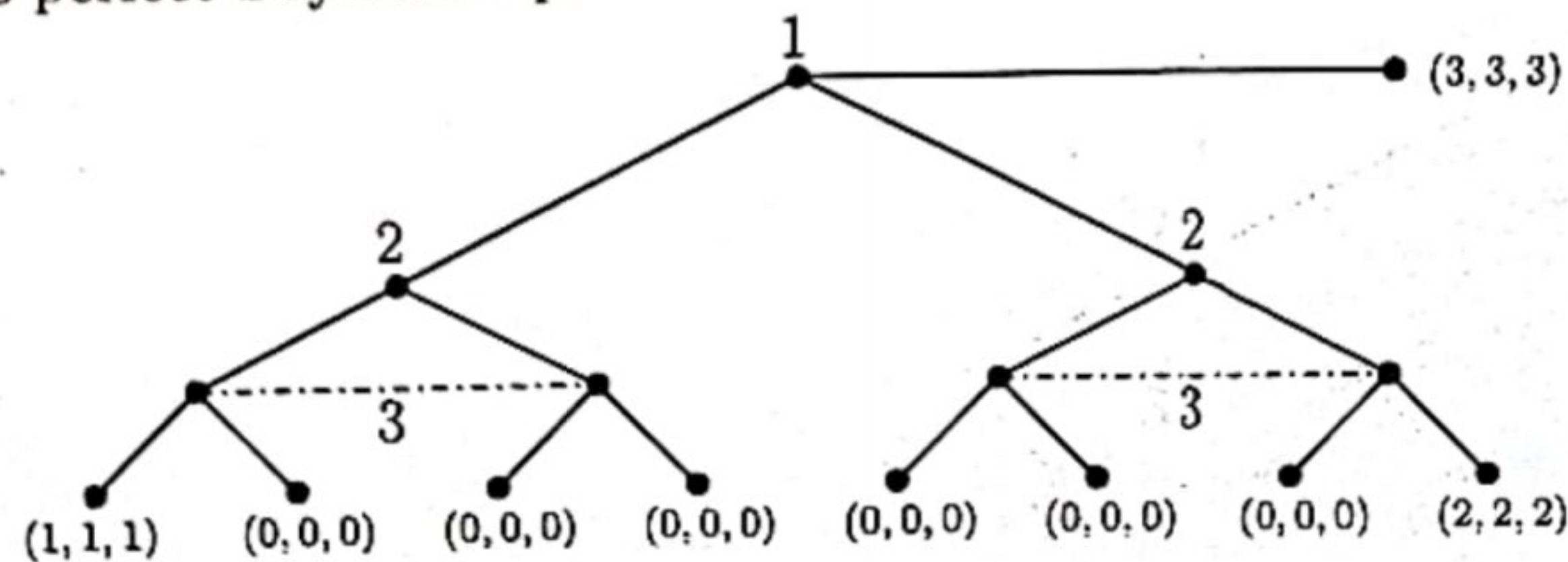
Quiz 2, Date: April 04, 2024, Thursday

Timing: 12:00 PM to 01:30 PM

All

- Answer any ~~four~~ questions from the following. The exam is for 20 marks.
- Try not to use any result not done in the class. However, if you use any such result, clearly state and prove it.
- Write your name, roll no., program name, and seat number clearly on the top of your answer sheet.
- For prove or disprove type questions, clearly state whether it's a prove or a disprove and then provide the arguments.
- One A4 sheet with ONLY necessary definitions and results are allowed during the exam. Use of a calculator, mobile, and smart watch is strictly prohibited.
- Be precise in writing the answers. Unnecessary arguments would lead to a deduction in marks.

1. Find all the perfect Bayesian equilibria for the following game. (7 marks)



2. Consider the following game (known as Nim game in the literature). It is a game between two-player where two piles of matches are placed before the players, and each pile has three matches. Each player in turn chooses a pile, and removes any number of matches from the pile he has selected (he must remove at least one match). The player who removes the last match wins the game. Prove or disprove that one of the players has a winning strategy in this game. If you prove the above statement, can you find out the winning strategy? (5 marks)

1 2
3. Allen and Rick need to divide \$100 between them as follows: first Allen suggests an integer x between 0 and 100 (which is the amount of money he wants for himself). Rick, on hearing the suggested amount, decides whether to accept or reject. If Rick accepts, the payoff of the game is $(x, 100 - x)$: Allen receives x dollars, and Rick receives $100 - x$ dollars. If Rick chooses to reject, neither player receives any money.

(i) Describe this situation as an extensive-form game. (2 marks)

(ii) What is the set of pure strategies each player has? (1 mark)

(iii) Show that any result $(a, 100 - a)$, $a \in \{0, 1, \dots, 100\}$, is a Nash equilibrium payoff. What are the corresponding equilibrium strategies? (2 marks)

(iv) Find all the subgame perfect equilibria (in pure strategies) of this game. (3 marks)





Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

MTH 636M: Game Theory

Quiz 3, Date: April 12, 2024, Friday

Timing: 06:15 PM to 07:30 PM

- Answer all the questions. The exam is for 20 marks.
- Try not to use any result not done in the class. However, if you use any such result, clearly state and prove it.
- Write your name, roll no., program name, and seat number clearly on the top of your answer sheet.
- For prove or disprove type questions, clearly state whether it's a prove or a disprove and then provide the arguments.
- One A4 sheet with ONLY necessary definitions and results are allowed during the exam. Use of a calculator, mobile, and smart watch is strictly prohibited.
- Be precise in writing the answers. Unnecessary arguments would lead to a deduction in marks.

1. Find all the max-min strategies and the value of the following two-player zero sum game: (5 marks)

		Player 2	
		a	b
Player 1	A	1	1
	B	2	1

2. Prove or disprove: The value of the two-player zero-sum game given by the matrix A is 0. Then the value of the two-player zero-sum game given by the matrix $-A$ is also 0. (5 marks)

3. A two-player game $G = (\{1, 2\}, (S_i)_{i \in \{1, 2\}}, (u_i)_{i \in \{1, 2\}})$ is called symmetric if

- (i) each player has the same set of strategies: $S_i = S_j$ for each $i, j \in N$, and
- (ii) $u_i(s_i, s_j) = u_j(s_j, s_i)$ for all $i, j \in \{1, 2\}$.

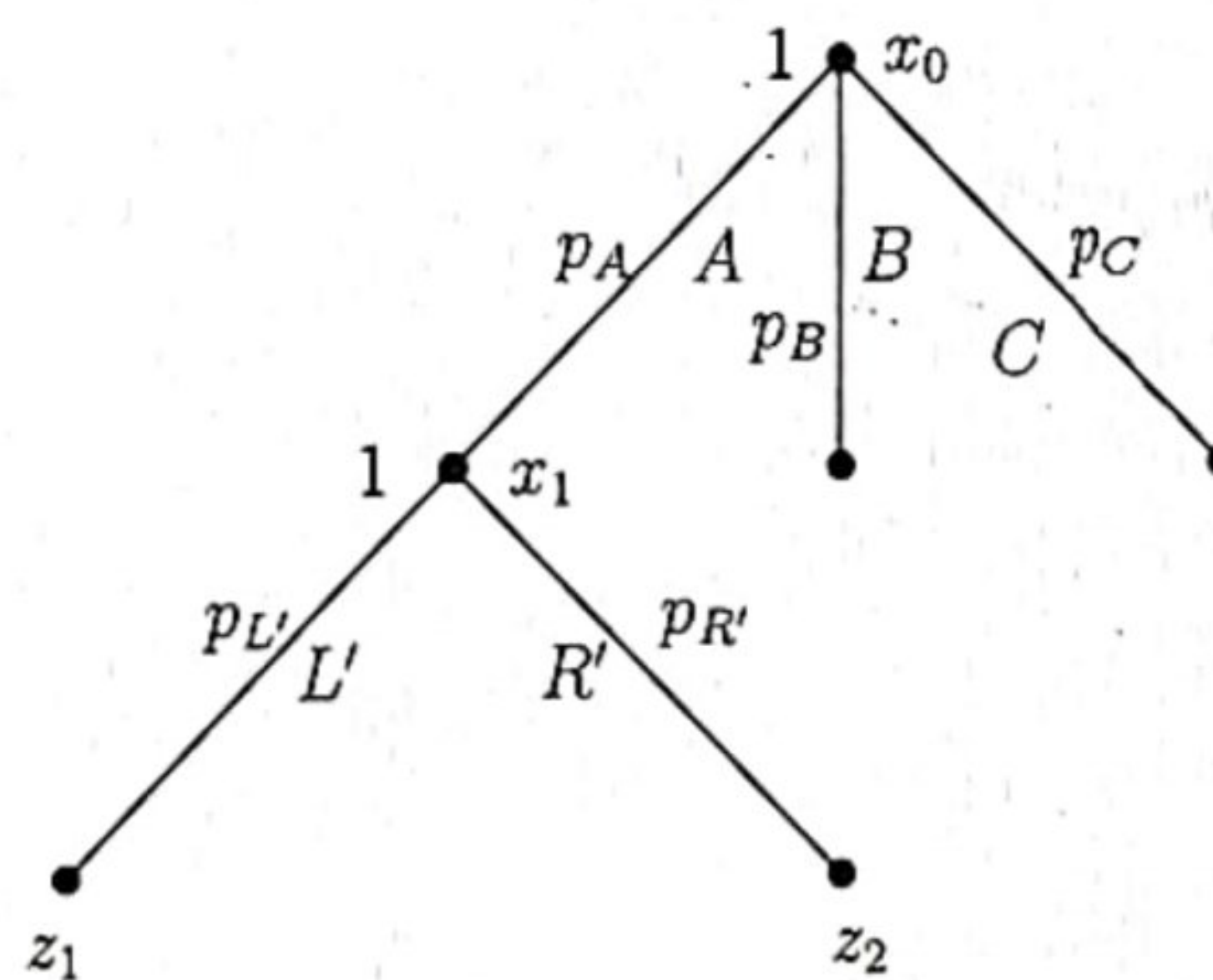
$$\begin{array}{cc|c} 0 & a_1 & \min \\ -a_2 & 0 & \\ \hline \max & 0 & a_1 \\ \hline & 0 & \end{array}$$

Prove or disprove: In every two-player symmetric game there exists a symmetric equilibrium in mixed strategies: an equilibrium $\sigma = (\sigma_i)_{i \in \{1,2\}}$ satisfying $\sigma_1 = \sigma_2$ (5 marks)

4. Determine all the behavioral strategies that are outcome equivalent with the following mixed strategy

AL'	AR'	BL'	BR'	CL'	CR'
0.5	0	0.2	0.1	0.1	0.1

in the following one-player extensive form structure:



(5 marks)



Indian Institute of Technology Kanpur

Department of Mathematics and Statistics

MTH 636A: Game Theory

End-Semester examination, Date: April 20, 2024,
Saturday

Timing: 5:00 PM to 8:00 PM

- This paper has seven questions. The exam is for 30 marks, and the maximum you can score is 37.
- Try not to use any result not done in the class. However, if you use any such result, clearly state and prove it.
- Write your name, roll no., program name, and seat number clearly in the appropriate place.
- For prove or disprove type questions, clearly state whether it's a prove or a disprove and then provide the arguments.
- One A4 sheet with ONLY necessary definitions and results is allowed during the exam. Use of a calculator, mobile, and smart watch is strictly prohibited.
- Be precise in writing the answers. Unnecessary arguments would lead to a deduction in marks.

1. Consider a game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ where S_i is finite for all $i \in N$. Suppose the players are pessimistic and the utility of player i for a mixed strategy profile σ is defined as follows:

$$\hat{u}_i(\sigma) = \min\{u_i(s) \mid s \in \text{Supp}(\sigma)\}$$

where $\text{Supp}(\sigma) = \{(s_1, \dots, s_n) \in S_1 \times \dots \times S_n \mid \sigma_i(s_i) > 0 \text{ for all } i \in N\}$. Either **prove or disprove** the following: The game G has a mixed strategy NE when players are pessimistic. (4 marks)

2. Find a Bayesian equilibrium in the following game with incomplete information:

- $N = \{I, II\}$,
- $T_I = \{I_1, I_2\}$ and $T_{II} = \{II_1\}$,
- $p(I_1, II_1) = \frac{2}{3}$, $p(I_2, II_1) = \frac{1}{3}$,
- Every player has two possible actions, and state games are given by the following matrices:

(B, T, R)

$1, 2, 3$
 $1, 3, 2$
 $2, 1, 3$
 $2, 3, 1$
 $3, 1, 2$
 $3, 2, 1$

		Player II	
		L	R
Player I	T	(2, 10)	(0, 3)
	B	(0, 4)	(1, 0)

Table 1: State game for $t = (I_1, II_1)$

		Player II	
		L	R
Player I	T	(0, 3)	(3, 1)
	B	(2, 0)	(0, 1)

Table 2: State game for $t = (I_2, II_1)$

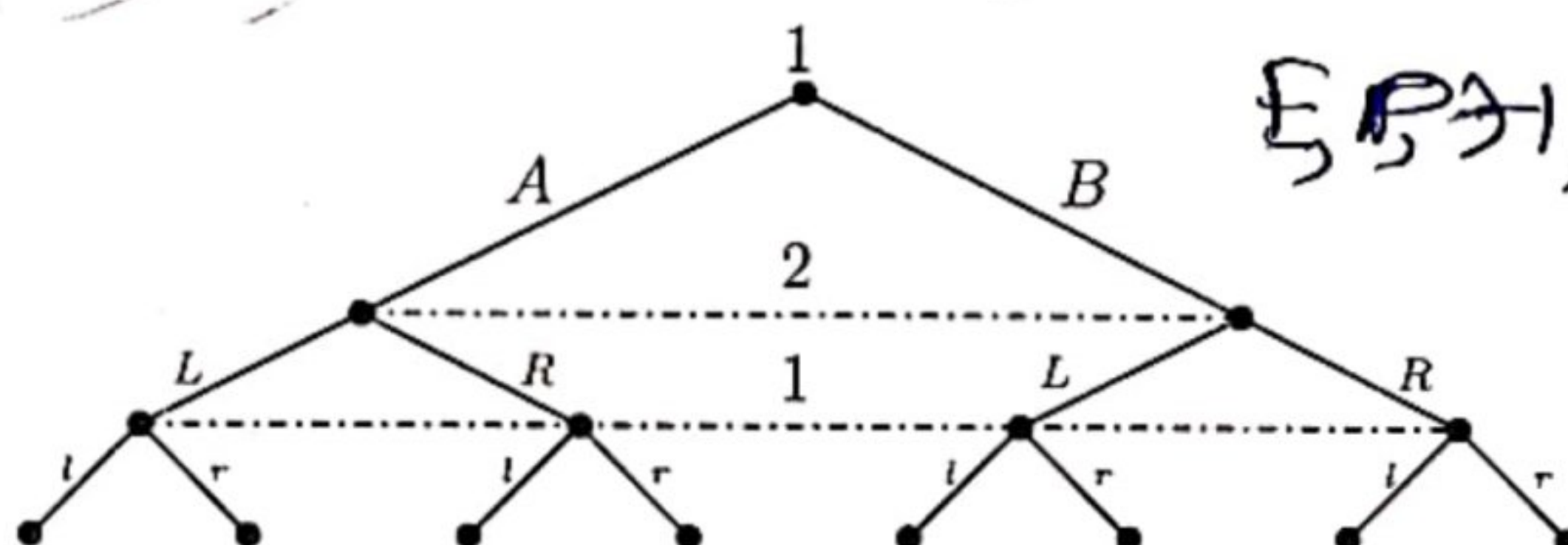
(I) $-1, -2, 3$
 (II) $-1, -2, 3$
 (III) $-1, -2, 3$

(5 marks)

3. A pack of three cards, labelled 1, 2, and 3, is shuffled. Sachin, Sourav, and Rahul each take a card from the pack. Each of the two players holding a card with low values (1 or 2) pays the amount of money appearing on the card he or she is holding to the player holding the high-valued card (namely, 3). Depict this situation as a game in extensive form. (5 marks)

B
 $= 3.2$

4. For the following game, **prove or disprove** that there is a mixed strategy that is not outcome equivalent to any behavioral strategies. (5 marks)



EPH, A, B, C

5. Consider the following game. Player I has the opening move, in which he chooses an action in the set $\{L, R\}$. A lottery is then conducted, with either λ or ρ selected, both with probability $\frac{1}{2}$. Finally, Player II chooses either l or r . The outcomes of the game are not specified. Depict the game tree associated with the extensive-form game in each of the following situations: (5 × 1.5 = 7.5 marks)

- Player II, at his turn, knows Player I's choice, but does not know the outcome of the lottery.
- Player II, at his turn, knows the outcome of the lottery, but does not know Player I's choice.
- Player II, at his turn, knows the outcome of the lottery only if Player I has selected L.

- (iv) Player II, at his turn, knows Player I's choice if the outcome of the lottery is λ , but does not know Player I's choice if the outcome of the lottery is ρ .
- (v) Player II, at his turn, does not know Player I's choice, and also does not know the outcome of the lottery.

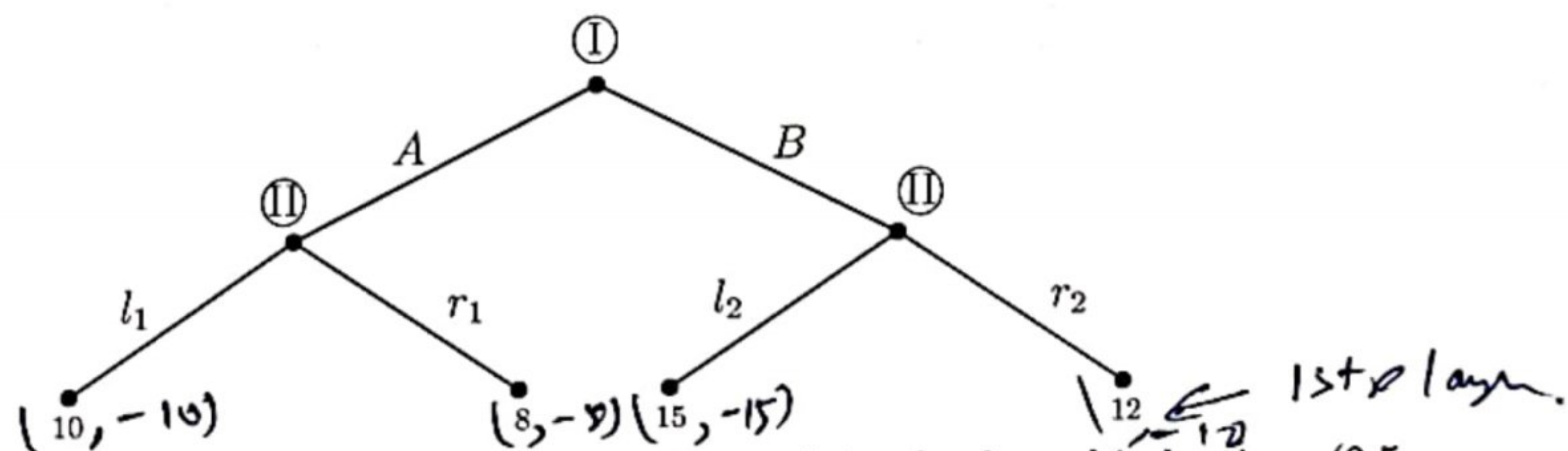
6. Establish whether there exists a two-player game in extensive form with perfect information, and possible outcomes I (Player I wins), II (Player II wins), and D (a draw), whose strategic-form description is

		Player II			
		s_{II}^1	s_{II}^2	s_{II}^3	s_{II}^4
Player I	s_I^1	D	D	D	D
	s_I^2	I	I	II	II
	s_I^3	II	II	D	D

If the answer is yes, describe the game. If not, explain why not.

(5 marks)

7. Consider the following two-player zero-sum game:



- (a) Find an optimal strategy for each player by applying backward induction. (0.5 marks)
- (b) Describe this game in strategic form. (1 marks)
- (c) Find all the optimal strategies of the two players. (2 marks)
- (d) Explain why there are optimal strategies in addition to the one you identified by backward induction. (2 marks)

Consider the following linear regression model

$$y = X_{n \times p} \beta + \epsilon, \quad (1)$$

where

- (i) $\beta = (\beta_0, \beta_1, \dots, \beta_{p-1})^T$ is a vector of real constants.
- (ii) $X_{n \times p}$ be a full column rank matrix with $p < n$ with first column a vector of all one's.
- (iii) $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T$ is the vector of errors such that ϵ_i are IID and $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ for $i = 1, \dots, n$.

Question 1:

- (1) Consider the regression model (1). Prove or disprove that $X_{n \times p} \beta = 0$ if and only if $\beta = 0$. (2 marks)
- (2) Consider the regression model (1). Suppose the hypothesis of interest is

$$H_0 : \beta_2 - \beta_5 = 0, \beta_3 - \beta_4 = 0 \text{ versus } H_1 : H_0 \text{ is not true.}$$

Provide an appropriate test statistics for testing the above hypothesis (do not derive) and find its distribution. (2 marks)

Question 2: Let μ be a vector of real constants of order $n \times 1$, I_n is the identity matrix of order $n \times n$, and $H = X_{n \times p} (X_{n \times p}^T X_{n \times p})^{-1} X_{n \times p}^T$. Prove or disprove that $\mu^T [I_n - H] \mu = 0$ if and only if μ belongs to the column space of $X_{n \times p}$. (3 marks)

Question 3: If $Y = (Y_1, Y_2)' \sim \mathcal{N}_2(0, \Sigma)$, where $\Sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}$, prove or disprove that

$$(Y' \Sigma^{-1} Y - \frac{Y_2^2}{\sigma_{22}}) \sim \chi_1^2. \quad (3 \text{ marks})$$

Question 1: Consider an experiment in which three treatment conditions (high, low and control) are tested on female rats. The objective of the study was to compare the birth weights of pups from litters (set of pups from the same mother) born to female rats that received the treatments.

(a) The above experiment has which type of data structure? (1 marks)

2f =

(b) Identify Level 1 and Level 2 observations in the above study. (1 marks)

(c) Let the following model is considered:

$$y_{ij} = \mu + \alpha_j + \beta_1 \text{Trt1}_j + \beta_2 \text{Trt2}_j + \epsilon_{ij}, \quad (1)$$

where

- (i) y_{ij} is the weight of i th pup from j th litter.
- (ii) μ , β_1 and β_2 are the fixed effects.
- (iii) The treatment effect for j th litter is coded as

$$(\text{Trt1}_j, \text{Trt2}_j) = \begin{cases} (0, 0), & \text{if treatment is Control} \\ (1, 0), & \text{if treatment is High} \\ (0, 1), & \text{if treatment is Low.} \end{cases}$$

(iv) α_j 's are identically and independently distributed (IID) as $\mathcal{N}(0, \sigma^2)$.

(v) The error ϵ_{ij} is independent of α_j and follows the following distribution

$$\epsilon_{ij} = \begin{cases} \text{IID } \mathcal{N}(0, \sigma_C^2), & \text{if treatment is Control} \\ \text{IID } \mathcal{N}(0, \sigma_H^2), & \text{if treatment is High} \\ \text{IID } \mathcal{N}(0, \sigma_L^2), & \text{if treatment is Low.} \end{cases}$$

Suppose there are 3 litters ($j = 1, 2, 3$) of size n ($i = 1, \dots, n$) each. Litters $j = 1$, $j = 2$, and $j = 3$ are receiving treatments Control, High, and Low respectively. Let the vector $\mathbf{y}_j$ is the vector of responses of n pups from the j th litter that is $\mathbf{y}_j = (y_{1j}, \dots, y_{nj})^T$ and $\mathbf{y} = (\mathbf{y}_1^T, \mathbf{y}_2^T, \mathbf{y}_3^T)^T$ where subscript T denotes the transpose of a vector.

Represent model (1) in a matrix form where the response vector is $\mathbf{y}$. (3 Marks)

2j

(d) Write the marginal model implied by model (1) in matrix form and find the variance covariance matrix of the response vector. (2+3 Marks)

Question 2: Consider an experiment conducted to examine nucleotide activation in three different brain regions among two adult male rats. There are two treatment conditions T1 and T2 are applied to each brain region and the responses are recorded after applying each treatment T1 and T2.

(a) The above experiment has which type of data structure? (1 marks)

(b) Identify Level 1 and Level 2 observations in the above study. (1 marks)

(c) Let the following model is considered:

$$y_{tij} = \underbrace{\mu}_{2\alpha} + \underbrace{\alpha_j + \beta_1 \text{RG1}_j + \beta_2 \text{RG2}_j}_{\times \beta} + \underbrace{\tau \text{Trt}_t}_{\alpha} + \underbrace{\epsilon_{tij}}_{\epsilon_{tij}} \quad (2)$$

where

- (i) y_{tij} is the response of j th rat from i th brain region for the treatment t where $i = 1, 2, 3$, $j = 1, 2$ and $t = 1, 2$.

(ii) μ , β_1 , β_2 and τ are the fixed effects.

(iii) The treatment effect is coded as

$$\tau_{\text{Trt}_t} = \begin{cases} 0, & \text{for treatment 1} \\ 1, & \text{for treatment 2.} \end{cases}$$

(iv) The region effect is coded as

$$(\text{RG1}_j, \text{RG2}_j) = \begin{cases} (0, 0), & \text{if } i = 1 \\ (1, 0), & \text{if } i = 2 \\ (0, 1), & \text{if } i = 3. \end{cases}$$

(v) α_j 's are IID $\mathcal{N}(0, \sigma^2)$.

(vi) The error ϵ_{ij} is independent of α_j and follows IID $\mathcal{N}(0, \sigma_\epsilon^2)$.

Let the response vector is $y = (\overbrace{y_{111}, y_{121}, y_{131}}, \overbrace{y_{112}, y_{122}, y_{132}}, \overbrace{y_{211}, y_{221}, y_{231}}, \overbrace{y_{212}, y_{222}, y_{232}})^T$. Represent model (2) in matrix form where the response vector is y . (3 Marks)

(d) Write the marginal model implied by model (2) in matrix form and find the variance covariance matrix of the response vector. (2+3 Marks)

Question 1: Consider the following simple linear regression model,

(1)

$$y_i = \beta_0 + x_i \beta_1 + \epsilon_i, \quad i = 1, \dots, n,$$

where errors ϵ_i 's are IID and follow $\mathcal{N}(0, \sigma^2)$.

(a) Write the MLE of σ^2 . (1 Marks)

Answer:

$$\hat{\sigma}^2 = \frac{(y - \hat{y})^T (y - \hat{y})}{n} \quad \text{or} \quad \frac{(y - X\hat{\beta})^T (y - X\hat{\beta})}{n} = \frac{y^T [I - H] y}{n}$$

where $H = X(X^T X)^{-1} X^T$

(b) Find $\sum_{i=1}^n \text{Var}(\hat{y}_i)$. (4 Marks)

Answer:

$$\sum_{i=1}^n \text{Var}(\hat{y}_i) = 2\sigma^2$$

(c) Find $\text{Cov}(\bar{y}, \hat{\beta}_1)$, where $\bar{y} = (1/n) \sum_{i=1}^n y_i$. (3 Marks)

Answer:

$$\text{Cov}(\bar{y}, \hat{\beta}_1) = 0$$

Question 2: Consider the following linear model,

(2)

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

where $i = 1, \dots, m$, $j = 1, \dots, n$, y_{ij} is the response of j th subject from i th cluster, μ is the fixed effect parameter, $\alpha_i \sim \mathcal{N}(0, \sigma_\alpha^2)$ IID and independent of ϵ_{ij} , and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ IID.

(a) Write the correlation $\text{Corr}(y_{ij}, y_{i'j'})$ where $j \neq j'$. (2 Marks)

Answer:

$$\text{Corr}(y_{ij}, y_{i'j'}) = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2}$$

(b) Write the model (2) in a matrix form where $y = (y_{11}, \dots, y_{1n}, \dots, y_{m1}, \dots, y_{mn})^T$ and all the random part should be included to the error term. (1 Marks)

Answer:

$$y = \mu \mathbf{1}_{mn \times 1} + \epsilon \begin{bmatrix} \epsilon_1 + \alpha_1 \\ \vdots \\ \epsilon_m + \alpha_m \end{bmatrix}$$

(c) Write the variance covariance matrix V of y . (3 Marks)

Answer:

$$\text{Var}(y) = V = \text{diag}(A, \dots, A)_{mn \times mn}$$

$$A = \begin{bmatrix} \sigma^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \dots & \sigma_\alpha^2 \\ \vdots & \ddots & \ddots & \vdots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \dots & \sigma^2 + \sigma_\alpha^2 \end{bmatrix}$$

- (d) Write the MLE of μ . Write the expression (do not solve it) in terms of $\hat{V}$ (estimate of V), y and appropriate design matrix. (2 Marks)

Answer:

$$\hat{\mu} = (1^T \hat{V}^{-1} 1)^{-1} 1^T \hat{V}^{-1} y$$

Question 3: Consider that $Y_i \sim \text{Poisson}(\mu_i)$ where $\mu_i = g(\eta_i)$; $\eta_i = \sum_{j=1}^p \beta_j x_{ij}$ and $i = 1, \dots, n$. Let the exponential link function $\mu_i = \exp(\eta_i)$ is used to relate the model parameters to the mean. Use the matrix notations to answer the following:

- (a) Find the likelihood equations (partial derivatives of likelihood function equated to 0). (3 marks)

Answer:

$$\sum_{i=1}^n (y_i - \mu_i) x_{ij} = 0 \quad \text{for } j = 1, \dots, p$$

comes 1 mark

- (b) Find the asymptotic variance matrix of ML estimator of $\beta = (\beta_1, \dots, \beta_p)'$. (3 marks)

Answer:

$$(X^T W X)^{-1} \quad \text{where } W = \text{Diag}(\mu_1, \dots, \mu_n)$$

$\mu_i = e^{\eta_i}$

Question 4: Consider the class of binary models with probability

$$\pi(x) = \Phi(\eta) \quad (3)$$

where $\eta = \alpha + \beta x$ and $\beta \neq 0$. Suppose that the standard cdf Φ corresponds to a probability density function ϕ that is symmetric around 0.

- (a) The value of x at which $\pi(x) = 0.5$ is (2 marks):

Answer:

$$x = -\alpha/\beta$$

- (b) The rate of change in $\pi(x)$ for the logit link ($\pi(x) = e^\eta / (1 + e^\eta)$) when $\pi(x) = 0.5$ is: (2 marks)

Answer:

$$\beta/4$$

- (c) The rate of change in $\pi(x)$ for the the probit link (defined in (3)) when $\pi(x) = 0.5$ is: (2 marks)

Answer:

$$\beta / \sqrt{2\pi}$$

Question 5: Consider the following nonlinear regression model

(4)

$$y_i = f(x_{1i}, x_{2i}, \theta_1, \theta_2) = x_{1i}\theta_1 + x_{2i}\theta_1\theta_2 + \epsilon_i, \text{ for } i = 1, \dots, n.$$

Let the true value of $\theta = \theta^* = (\theta_1^*, \theta_2^*)^T = (1, 1)^T$ and $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ and IID. Also, let $x_1 = (x_{11}, \dots, x_{1n})^T$ and $x_2 = (x_{21}, \dots, x_{2n})^T$ are such that $x_1^T x_1 = 1$, $x_2^T x_2 = 1$, $x_1^T x_2 = 0$, $y^T x_2 = 0$ and $x_1^T x_2 = 0$. Suppose $\hat{\theta}$ denotes the least-square-estimator of θ . Then

(a) Find the asymptotic distribution of $\hat{\theta}$. (3 marks)

Answer:

$$\hat{\theta} \sim \mathcal{N}(\theta^*, \sigma^2 C^{-1}) \quad C = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad \theta^* = (1, 1)^T$$

$$C^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

(b) Find the variance and covariance of $\hat{\theta}_1$ and $\hat{\theta}_2$. (1 mark)

Answer:

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \quad \text{Var}(\hat{\beta}_2) = 2\sigma^2 \quad \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) = -\sigma^2$$

(c) What is δ (step size) in the numerical algorithms to find LSE of θ when a linear approximation of $f(\cdot)$ is used. Write your answer only in terms of (θ_1, θ_2) . (4 marks)

Answer:

$$f^{(4)} = \begin{bmatrix} -\theta_1^{(4)} \\ 0 \end{bmatrix}$$

(d) Find a $(1 - \alpha)\%$ confidence interval for $\theta_1 - \theta_2$. Write your answer only as a function of $(\hat{\theta}_1, \hat{\theta}_2)$, critical value of an appropriate distribution and an unbiased estimator of σ^2 . (3 marks)

Answer:

$$(\hat{\theta}_1 - \hat{\theta}_2) \pm t_{n-2}^{1/2} s \sqrt{a^T \hat{\sigma} a} \quad a = [1, -1]^T; \quad s^2 = \frac{(y - f(\hat{\theta}))^T (y - f(\hat{\theta}))}{n-2}$$

$$(\hat{\theta}_1 - \hat{\theta}_2) \pm t_{n-2}^{1/2} s \frac{1}{\hat{\theta}_2} [1 + (\hat{\theta}_1 + \hat{\theta}_2)^2]^{1/2}$$

(e) Find a $(1 - \alpha)\%$ confidence interval for $a(\theta) = \theta_1\theta_2$. Write your answer only as a function of $(\hat{\theta}_1, \hat{\theta}_2)$, critical value of an appropriate distribution and an unbiased estimator of σ^2 . (3 marks)

Answer:

$$\hat{\theta}_1 \hat{\theta}_2 \pm t_{n-2}^{1/2} s$$

Question 6: Consider the following linear regression models

(5)

$$y_i = x_i^T \beta_1 + \epsilon_{1i}, \quad i = 1, \dots, k,$$

(6)

$$y_i = x_i^T \beta_2 + \epsilon_{2i}, \quad i = k+1, \dots, n,$$

where β_d is a $p \times 1$ ($d = 1, 2$), errors ϵ_{di} are independently distributed as $\mathcal{N}(0, \sigma_d^2)$ for $d = 1, 2$. In matrix form, the above model can be expressed as

$$y_1^k = X_1^k \beta_1 + \epsilon_1^k, \quad (7)$$

$$y_2^k = X_2^k \beta_2 + \epsilon_2^k, \quad (8)$$

where y_1^k is $k \times 1$ and y_2^k is $(n - k) \times 1$. Here $(\beta_1, \beta_2, \sigma_1^2, \sigma_2^2, k)$ are unknown parameters.

(a) Find the MLE $\hat{\beta}_1$ of β_1 . (1 marks)

Answer:

$$\hat{\beta}_1 = (X_1^k X_1^k)^{-1} X_1^k y_1^k$$

(b) Find the MLE $\hat{\beta}_2$ of β_2 . (1 marks)

Answer:

$$\hat{\beta}_2 = (X_2^k X_2^k)^{-1} X_2^k y_2^k$$

(c) Find the MLE $\hat{\sigma}_1^2$ of σ_1^2 . (1 marks)

Answer:

$$\hat{\sigma}_1^2 = \frac{\hat{S}_1^k}{k}, \quad \hat{S}_1^k = \sum_{i=1}^k (y_1^k - x_1^k \hat{\beta}_1)^2$$

(d) Find the MLE $\hat{\sigma}_2^2$ of σ_2^2 . (1 marks)

Answer:

$$\hat{\sigma}_2^2 = \frac{\hat{S}_2^k}{n-k}, \quad \hat{S}_2^k = \sum_{i=1}^k (y_2^k - x_2^k \hat{\beta}_2)^2$$

(e) Find the MLE $\hat{\sigma}^2$ of σ^2 when $\sigma_1^2 = \sigma_2^2 = \sigma^2$. (2 marks)

Answer:

$$\hat{\sigma}^2 = \frac{\hat{S}_1^k + \hat{S}_2^k}{n}$$

(f) Write the function $M(k)$ which is maximised to obtain the MLE of k when $\sigma_1^2 = \sigma_2^2 = \sigma^2$. (2 marks)

Answer:

$$M(k) = -\frac{n}{2} - \frac{n}{2} \log(2\pi) - \frac{n}{2} \log\left(\frac{\hat{S}_1^k + \hat{S}_2^k}{n}\right)$$

If write only this, full marks.

Quiz 1 - MTH 314A 2024**MULTIVARIATE ANALYSIS**

Time 15 Minutes, M.M. 10

Let a p -component column vector $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$ is partitioned as $\underline{X} = (\underline{X}_1 \quad \underline{X}_2)'$ with $p/2$ components in $\underline{X}_1$ and $p/2$ components in $\underline{X}_2$. Also, $\underline{\mu}$ and Σ are similarly partitioned as $\underline{\mu} = (\underline{\mu}_1 \quad \underline{\mu}_2)'$ and $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$. Find the (i) distribution of $(\underline{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underline{X}_2)$ (ii) covariance matrix between $\underline{X}_1$ and $(\underline{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underline{X}_2)$ and (iii) covariance matrix between $(\underline{X}_2 - \Sigma_{21}\Sigma_{11}^{-1}\underline{X}_1)$ and $(\underline{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underline{X}_2)$.

Time: 15 Minutes

M.M.:10

QUIZ 2- MTH-314 MULTIVARIATE ANALYSIS

Name:

Roll No.

Let the p -component column vectors $\underline{X}_\alpha \sim N_p(\underline{\mu}, \Sigma), \alpha = 1, 2, \dots, N$ are independently distributed. What is the

distribution of the matrix $S = \frac{\sum_{\alpha=1}^N (\underline{X}_\alpha - \bar{\underline{X}})(\underline{X}_\alpha - \bar{\underline{X}})'}{N-1}$ where all the symbols have the usual meaning? You can

directly use the information about the distribution of $\sum_{\alpha=1}^N (\underline{X}_\alpha - \bar{\underline{X}})(\underline{X}_\alpha - \bar{\underline{X}})'$.

MID SEM EXAMINATION-2024
MTH-314 MULTIVARIATE ANALYSIS

Time: 2 Hours

M.M.: 35

Attempt all questions. All symbols have the usual meaning.

1. Prove that if $\underline{X}_1 \sim N_r(\underline{\mu}_1, \Sigma_{11})$ and $(X_2 | X_1 = x_1) \sim N_{p-r}(Ax_1 + b, \Omega)$ where Ω does not depend on x_1 , then $\underline{X} = \begin{pmatrix} \underline{X}_1 \\ \underline{X}_2 \end{pmatrix} \sim N_p(\underline{\mu}, \Sigma)$ where $\underline{\mu} = \begin{pmatrix} \underline{\mu}_1 \\ A\underline{\mu}_1 + b \end{pmatrix}$ and $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{11}A' \\ A\Sigma_{11} & \Omega + A\Sigma_{11}A' \end{pmatrix}$. [10]

2. If the sample covariance matrix A of order $p \times p$ is distributed as $W(\Sigma, n)$, C is a nonsingular nonstochastic matrix of order $p \times p$, then derive the characteristic function of $C'AC$. Here $A = \sum_{\alpha=1}^n \underline{Y}_\alpha \underline{Y}_\alpha'$ where $\underline{Y}_\alpha \sim N(0, \Sigma)$, $\alpha = 1, 2, \dots, n$ are independently distributed. [10]

3. If $A \sim W(\Sigma, n)$, then find the probability distribution of A^{-1} where all the symbols have the usual meaning. [5]

4. Consider an i.i.d. sample of size 5 from a bivariate normal distribution $\underline{X} \sim N_2\left(\underline{\mu}, \begin{pmatrix} 3 & \rho \\ \rho & 1 \end{pmatrix}\right)$

where ρ is a known parameter. Suppose $\underline{x}' = (1, 0)$. For what value of ρ would $H_0: \mu' = (0, 0)$ be rejected in favour of $H_1: \mu' \neq (0, 0)$ at 5% level of significance. For simplicity in the calculation, consider the critical value (details to be stated by you) at 5% level of significance as 2. [5]

5. A sample of n paired observations on $\underline{x}_i$ and $\underline{y}_i, i = 1, 2, \dots, n$ are obtained. Assume that $\underline{X}_{p \times 1}$ and $\underline{Y}_{p \times 1}$ are correlated and $\begin{pmatrix} \underline{X} \\ \underline{Y} \end{pmatrix} \sim N_{2p}\left(\begin{pmatrix} \underline{\mu}_x \\ \underline{\mu}_y \end{pmatrix}, \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}\right)$ with unknown $\Sigma_{xx}, \Sigma_{xy}, \Sigma_{yx}$ and Σ_{yy} . How would you test $H_0: \underline{\mu}_x = \underline{\mu}_y$. Explain, develop, and describe the test statistic. [5]

**MTH 314 - MULTIVARIATE ANALYSIS
END SEMESTER EXAMINATION-2024**

Time: 3 hours

MM: 40+20

Note: Attempt all questions. All symbols have the usual meaning. The second part of the paper in page numbers 2-6 is to be solved on the question paper itself and is to be submitted.

- 1 Consider the distribution of the set of correlation coefficients when the population covariance matrix is a diagonal matrix with different elements. Consider the case when $N = 5$ and $p = 2$. If $P(|r| \geq K) = \alpha$ then show that K is the root of the equation $K\sqrt{1-K^2} + \sin^{-1}K - \frac{\pi(1-\alpha)}{2} = 0$. [10]

- 2 Let $\underline{X}_\alpha \sim N_p(\underline{\mu}, \Sigma)$, $\alpha = 1 \dots N$ ($N > p$).

- (a) Suppose the data on $\underline{X}_\alpha$ is standardized using the transformation as $\underline{Z} = \Sigma^{-1/2} \underline{X}$. Would it be reasonable to apply the principal component analysis on the standardized data? Justify your answer with reason. $\text{Cov}(\underline{Z} \underline{Z}') = \Sigma^{-1/2} \underline{X} \underline{X}' \Sigma^{-1/2}$ [No marks without correct reason] [3]

- (b) Suppose the covariance matrix of a two-dimension random vector is $\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$ where ρ is a known parameter. Find out all the principal components and their variances. $\Sigma^{-1/2} \Sigma^{1/2} \Sigma^{-1/2} \Sigma^{1/2} = \underline{I}$

3. Let $(\underline{X}_1, \underline{X}_2, \dots, \underline{X}_n)$ be a random sample from $N(0, \Sigma)$. Each $\underline{X}_i$ is partitioned into two components with p_1 and p_2 number of variables with $p_1 < p_2$. An experimenter obtains the canonical variables $(U_1, V_1), \dots, (U_{p_1}, V_{p_1})$. Another experimenter obtains the canonical variables $(kU_1, kV_1), \dots, (kU_{p_1}, kV_{p_1})$ where $k > 0$ is a scalar.

- (a) What will be the difference in the analysis based on the two sets of canonical variables? Explain with reason. [No marks without correct reason]. [3]

- (b) Derive the first and second sets of canonical variables based on $(kU_1, kV_1), \dots, (kU_{p_1}, kV_{p_1})$. [7]

4. (a) Calculate the maximum likelihood discrimination rule based on one observation from a one-dimensional variable with an exponential distribution with p.d.f. $f(x) = \lambda \exp(-\lambda x)$ for $x > 0$.

[Hint: $\pi_1 : \text{Exp}(\lambda_1)$, $\pi_2 : \text{Exp}(\lambda_2)$]

[5]

- (b) Derive a discriminant rule for classifying an observation into one of the two populations (π_1 and π_2)

minimizing the expected cost of misclassification considering the prior probability $q_1 = \frac{1}{3}$ and expected cost of misclassification $C(2/1) = 2C(1/2)$. [5]

$$\lambda = \alpha_1' \Sigma^{-1} \alpha_2$$

$$(\Sigma - \lambda \underline{I}) \alpha = 0$$

Page 1 of 5

$$A \alpha = \lambda \alpha$$

$$\Sigma \alpha = \lambda \alpha$$

$$\alpha^{(2)'} \Sigma \alpha^{(2)} = \alpha^{(2)'} \lambda \alpha^{(2)} \rightarrow \lambda \alpha^{(2)'} \alpha^{(2)} = 1$$



Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
An Introduction To Bayesian Analysis (MTH422A)

End Semester Examination
Date: April 24, 2024, Wednesday

Timing: 12:30 AM – 2:30 PM (2 hours)

Spring 2024

Max point: 25

The Weibull distribution has a density function $f(x) = \alpha\beta x^{\alpha-1} \exp[-\beta x^\alpha]$ for $x \in \mathbb{R}^+$. Here the shape and rate parameters are given by $\alpha > 0$ and $\beta > 0$. Suppose $X_1, \dots, X_n$ denote the annual total rainfall at IITK Airstrip for n years. You model the data as IID observations (assuming “our tolerance and habits have changed but the climate didn’t”) from a Weibull distribution with known shape $\alpha = \alpha_0$ (unless explicitly mentioned that it is unknown as in Q7) but assume that β is unknown. If you cannot solve Q1, then I will tell you the answer. In that case your marks in Q1 will be zero but it will save marks in other questions. If answer to Q1 is wrong, the most of the other answers will be wrong and so be careful. In Q7 and Q8, you have to explicitly mention the steps like initialization, updates, discarding undesired samples, etc. up to the information asked in the questions. For each question where the answer is a single number or expression, put a box around your answer to highlight.

1. Choose a conjugate prior for β with generic choice of hyperparameters (i.e., if it is a normal distribution, choose its mean and variance to be some m and s^2 but don’t fix $m = 0, s = 100$). Show that your choice is indeed a conjugate prior. No point will be given if you don’t show. (2 points)
2. Calculate the posterior mean, variance, 95% equal-tailed exact credible interval, and 95% Gaussian-approximated asymptotic credible interval for β . (1+1+1+1=4 points)
3. Prediction of the amount of rainfall for the $(n+1)$ -th year is necessary for agricultural experiments going on at IITK. Here, not only the predicted value is important but also the whole predictive distribution is important for accurate uncertainty quantification. Derive the analytical expression of the posterior predictive density function. (2 points)
4. We want to decide between two models

$$\mathcal{M}_1 : X_i \stackrel{\text{IID}}{\sim} \text{Weibull}(\alpha_0, \beta_0), \quad \mathcal{M}_2 : X_i | \beta \stackrel{\text{IID}}{\sim} \text{Weibull}(\alpha_0, \beta), \quad \beta \sim \pi(\beta),$$

where $\pi(\beta)$ is as in Q1. Find the Bayes factor $BF_{2,1}$ in terms of $n, \alpha_0, \beta_0, X_i$ ’s and the hyperparameters in $\pi(\beta)$. (2 points)

5. Two faculties from the Department of Earth Sciences are unhappy about the non-uniqueness of the final inferences due to the choice of priors. Profs. A and B want to choose priors $\pi_A(g(\beta))$ and $\pi_B(h(\beta))$ for two different transformations of β given by $g(\beta)$ and $h(\beta)$ respectively, but they want to ensure that both of them end up with the same posterior for β . Derive a prior for β that ensures this condition. (2 points)
6. The mean of a $\text{Weibull}(\alpha_0, \beta)$ distributed random variable is given by $\Gamma(1 + \alpha_0^{-1})\beta^{-1}$. Hence, we are interested in summarizing the posterior of β through a single measure that minimizes the posterior risk under the loss function $\ell(\hat{\beta}, \beta) = [\hat{\beta}^{-1} - \beta^{-1}]^2$. Find the Bayes rule. (2 points)

7. In practice, we never know α_0 and so one option is to treat α as unknown along with β . Here no known conjugate prior exists for α . To draw Bayesian inference, we now need to draw samples from the posterior distribution $\pi(\alpha, \beta | X_1, \dots, X_n)$. Write down a Metropolis-Hastings within Gibbs sampling scheme here. How will you use these samples to further approximate the posterior predictive density? (3+1 = 4 points)
8. Prof. C from the Department of Earth Sciences does not believe that the climate didn't change and hence he prefers to check whether there is any trend in the data or not. However, due to his limited knowledge in generalized linear models, he transform the data into the log-scale $Y_i = \log[X_i]$, $i = 1, \dots, n$ and fit a model $Y_i | \beta_0, \beta_1, \sigma^2 \stackrel{\text{Indep}}{\sim} \text{Normal}(\beta_0 + \beta_1 Z_i, \sigma^2)$, where Z_i 's are some known transformed forms of $i = 1, \dots, n$. Hence, if β_1 is significantly away from zero, we would conclude that there is a trend in the data across years. He wants to perform Bayesian LASSO. Write a Gibbs sampling algorithm that does not involve any Metropolis-Hastings step or any accept-reject-type step. Choose priors properly and mention the full conditional distributions of parameters and hyperparameters. You don't have to derive the parameters of the full conditional distributions explicitly but just mention the names of those distributions only. (3 points)
9. Prof. C found that zero lies within the 95% credible interval of β_1 and hence he concludes that there is no significant linear trend in log-rainfall. You are not convinced because Bayesian LASSO shrinks the posterior distribution of β_1 towards zero. You want to compare the results with a simple Bayesian regression model with noninformative priors. Mention four ways (just the names) of doing this. (2 points)
10. Unobservable climate conditions can possibly induce dependence in the observations. Hence, Prof. E from the Department of Computer Science and Engineering wants to explore a hidden Markov model as follows:

$$Y_i | W_i, \beta_0, \beta_1, \sigma^2 \stackrel{\text{Indep}}{\sim} \text{Normal}(\beta_0 + \beta_1 Z_i + W_i, \sigma^2)$$

$$W_1 \sim \text{Normal}(0, \tau^2), W_i | W_{i-1} \sim \text{Normal}(\rho W_{i-1}, \tau^2(1 - \rho^2)), i = 2, \dots, n$$

$$\beta_0, \beta_1 \stackrel{\text{IID}}{\sim} \text{Normal}(0, 100^2), \sigma^2, \tau^2 \stackrel{\text{IID}}{\sim} \text{Inverse-gamma}(0.01, 0.01), \rho \sim \text{Uniform}(-1, 1)$$

Write down a directed acyclic graph (DAG) representation of the above model. (2 points)

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 676; ECONOMETRICS

(Session: 2023-24)

Semester: II)

Quiz 2

Total Marks: 20

Time: 45 minutes

1. The following regression results were obtained from the Consumer Price Index (CPI) data of India (Y) for the period 1985 to 2021:

$$\Delta \bar{Y}_t = 31.03 - 0.188Y_{t-1}$$

Std. error (12.50) (0.080)

Check the stationarity of the Indian CPI series using the (a) DF test and (b) usual t-test, at 5% level of significance.

(Write clearly the null and alternative hypotheses, the test statistic and its observed value, critical region, critical values from table for comparison - interpolate, if necessary, and the final verdict about nature of the series.) [6 marks]

Table of critical values of DF and t-tests, α = area in the lower (left) tail

	DF test		t-test	
Sample Size(T)	$\alpha = 0.025$	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.05$
25	-2.88	-3.00	-2.06	-1.71
50	-2.84	-2.93	-2.01	-1.68
100	-2.81	-2.89	-1.98	-1.66

2. (a) Expand the following rational lag model as an IDL model, and obtain the short and long run effects of the regressor (X) on the response (Y):

$$Y_t = [(0.6 + 2L)/(1 - 0.6L + 0.5L^2)]x_t + \epsilon_t.$$

- (b) Obtain the ARDL form of the following model:

$$Y_t = \alpha + \frac{\beta + \gamma L}{1 - \delta_1 L - \delta_2 L^2} x_t + \epsilon_t.$$

Are OLS estimates of the ARDL model parameters consistent if (i) $\epsilon_t \sim WN(0, \sigma^2)$ and (ii) $\epsilon_t \sim$ stationary AR(2) process? Give reasons for your answer. [4 + 4 = 8 marks]

3. Consider the MLR model $Y_{T \times 1} = X\beta_{k \times 1} + \epsilon$, suppose $p \lim_{T \rightarrow \infty} X'\epsilon \neq 0$.

- (a) Define the IV estimator of β in a just identified case and stating all necessary assumptions, prove that this estimator is consistent.

- (b) What would the estimator be in an over identified case? [3 + 3 = 6 marks]

Handwritten calculations and notes:

- Calculation of $1/10 = 0.1$
- Calculation of $1.188(235)$ with intermediate steps: $163, 28, 24, 0.40$
- Calculation of 1.19 with intermediate steps: $1.71, 1.68, 1.39, 1.19$
- Calculation of 21021 with intermediate steps: $1985, 0036$
- Calculation of 85 with intermediate steps: $16, 24, 32$
- Calculation of $34/100$
- Calculation of $6 \times 10/10$
- Calculation of $3/10$
- Calculation of 1.016×0.6 with intermediate steps: $0.216, 0.060, 0.0216$
- Calculation of $0.6 \times 1.016 - 0.5 \times 2.36$ with intermediate steps: $2.36 \times 0.5, 1.18, 10.50$
- Calculation of 1.016×0.6 with intermediate steps: $0.216, 0.060, 0.0216$
- Calculation of $0.6 + 2.36 + 1.016$

DEPARTMENT OF MATHEMATICS & STATISTICS
MTH 676: ECONOMETRICS

(Session: 2023-24

Semester: II)

End-Semester Exam

Total Marks: 90

Time: 180 minutes

1(a) Consider a system of multivariate regression equations of N endogenous variables with k explanatory variables for T time-points in each equation. Random errors across equations are contemporaneously correlated, but within equation errors are uncorrelated and homoscedastic.

(i) Obtain estimates of the model parameters in the above set-up.

(ii) Derive expectation and variance-covariance of the estimators.

(b) An economist proposes a model where Inflation Rate (INF) and Exchange Rate (FX) of India are explained by its Net Exports (NX) and the International Oil Price (IOP). She claims that the “marginal effect of IOP on INF is same as that of NX on FX ”. Summary from monthly data of the variables for the period Apr-2016 to May-2020, is given below:

(i) Fit the multivariate MLR model with the given data.

(ii) Assuming the asymptotic normality of the relevant test statistic/s, test the hypothesis, in detail, and verify whether the economist is justified in her claim. [10 + 8 = 18]

Sum of squares and sum of products

	IOP	NX	INF	FX	Critical values
IOP	50				$\Phi(1.64) = 0.95$
NX	50	100			
INF	100	75	250		$\Phi(1.96) = 0.975$
FX	50	150	100	500	

$$\Phi(z) = P(Z \leq z) \text{ for } Z \sim N(0,1)$$

2. An economist uses a standard VAR(1) (with intercept) for $Y = \text{INR_USD Exchange Rate}$ and $X = \text{Import Price Index}$ over a sample of T time points. The error follows $N_2(0, \Sigma)$ distribution.

(a) What are the BLUEs of the coefficients? What are estimates of the elements of the error variance-covariance matrix Σ ?

(b) How would you test the hypothesis “INR_USD Rate does not Granger cause Import Price”?

(c) Derive the parameters of the ARDL(1,1) model $Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$ from the parameters of the VAR. Obtain $E(\varepsilon_t)$ and $Var(\varepsilon_t)$.

(d) Suppose y_t and x_t are $I(1)$ and $z_t = y_t - \beta x_t$ is $I(0)$. What is the associated vector ECM? What is the relationship among the parameters of the VECM and the VAR? [4 + 4 + 6 + 4 = 18]

3(a) Explain through an example why the Engel-Granger single equation model may give inconclusive results about cointegration among a group of three or more variables.

(b) Explain, in detail, Johanssen's tests of cointegration in a VAR setup.

(c) Let $X_{it} = \mu_{it} + \epsilon_{it}$, $i = 1, 2, 3$ be three trending variables where μ_{it} is the stochastic trend and ϵ_{it} is the white noise. What is the condition on the three variables so that they are cointegrated? Hence obtain two unique cointegrating relationship vectors for the three variables. How would you modify your answer if the variables have a common stochastic trend, say, μ_t . [3 + 8 + 7 = 18]

4. Consider the following model where Consumption (C) and Investment (I) are jointly determined by the two equations: $C = \beta_0 + \beta_1 I + \beta_2 Y + \epsilon_C$ and $I = \alpha_0 + \alpha_1 C + \epsilon_I$. All variables are observed for T time points. ϵ_C and ϵ_I are classical errors with usual assumptions, but contemporaneously correlated across equations. Income (Y) is uncorrelated with the errors.

(a) Show that the OLS estimate of α_1 from the investment equation is inconsistent. Under which condition/s, is it consistent?

(b) Identify a valid instrument in the system; stating necessary regularity conditions on this instrument, obtain the corresponding IV estimate of α_1 and prove that it is consistent.

(c) Obtain the 2SLS estimate of α_1 and show that it is the OLS estimate obtained in part (b).

(d) Suppose $\alpha_0 = 0$. Is $\bar{I}/\bar{C}$ a consistent estimate of α_1 ? Justify your answer.

(Note: All answers should be in terms of the model-specific variables Y, C, I , etc.) [4 + 6 + 6 + 4 = 20]

5. Consider the "Model of Income Determination" for jointly determining consumption (C), investment (I) and income (Y):

$$C_t = \alpha_0 + \alpha_1 y_t + \alpha_2 C_{t-1} + \epsilon_{1t}$$

$$I_t = \beta_0 + \beta_1 r_t + \beta_2 (y_t - y_{t-1}) + \epsilon_{2t}$$

$$Y_t = C_t + I_t + G_t$$

$$\frac{\text{Var}(p)}{\gamma_{t-1} + \dots + \gamma_{t-p}}$$

(a) What is the completeness condition of the above model?

(b) Obtain the reduced form of the model.

(c) How many extra conditions is required to estimate parameters of structural form of the model?

(d) What are the rank and order conditions for identification of the model? [4 + 8 + 2 + 2 = 16]

Bayesian Econometrics (ECO545A)
Semester II 2023-24
Midterm – Total Points: 100

Instructor: M.A. Rahman

February 19, 2024

Please read the instructions carefully and follow them while writing answers.

- Questions should be answered in order as they appear in the exam. Every new question should begin in a new page.
- Full points will be awarded only if the answers are correct and to the point.

1. (10+10=20 points). Consider a random sample of size n from a $N(\theta, \sigma^2)$ distribution with σ^2 known. Suppose this yields the 95% confidence interval $CI(8 < \theta < 12)$. Carefully interpret this interval.

Now suppose you use the prior $\theta \sim N(0, 10)$ with the above data and obtain the posterior distribution $\theta|y \sim N(10, 1.44)$. Suppose, the 95% posterior probability interval (aka credible interval) comes out to be $PI(7 < \theta < 13)$. What is the interpretation of this interval? Why is the interpretation of this interval different, if at all, from that obtained via Classical approach?

2. (10+10=20 points). Consider the weighted error loss function

$$c(\hat{\theta}, \theta) = w(\theta)(\theta - \hat{\theta})^2, \quad w(\theta) > 0.$$

Show that the Bayesian point estimate corresponding to $c(\hat{\theta}, \theta)$ is given by

$$\hat{\theta} = \frac{E_{\theta|y}[\theta w(\theta)]}{E_{\theta|y}[w(\theta)]}$$

3. Explain in detail the role of $w(\theta)$ to that of θ .

Recall, Bayesian point estimate minimizes the expected posterior loss i.e., minimizes the expected loss function where the expectation is taken with respect to the posterior distribution of the parameter.

3. (30 points). Suppose Y is a Bernoulli random variable with p.m.f. $\Pr(Y = \theta - 1|\theta) = \Pr(Y = \theta + 1|\theta) = 0.5$ and $\theta \in \mathbb{R}$. Suppose the prior distribution is $\theta \sim N(0, 1)$ with prior density $\phi(\cdot)$. Given one observation $Y = y$, find the posterior distribution of θ .

4. (10+10+10 = 30 points). Consider the regression model,

$$y_i = x_i' \beta + \epsilon_i, \quad i = 1, \dots, n,$$

and assume that ϵ_i has the asymmetric Laplace (AL) distribution with density,

$$f(\epsilon_i) = p(1-p) \exp[-\rho_p(\epsilon_i)], \quad \checkmark$$

where $\rho_p(\epsilon_i) = \epsilon_i \{p - I(\epsilon_i < 0)\}$, p denotes the quantile, and $I(\cdot)$ is an indicator function which takes the value 1, if the condition within parenthesis is true and 0 otherwise.

The above layout is that of Bayesian linear quantile regression. However, the given formulation does not yield a convenient MCMC algorithm. Consequently, the error ϵ is expressed as a mixture of a normal-exponential distribution as follows:

$$\epsilon_i = \theta w_i + \tau \sqrt{w_i} u_i, \quad \checkmark$$

where w_i and u_i are mutually independent, $u_i \sim N(0, 1)$, $w_i \sim \mathcal{E}(1)$, and $\mathcal{E}$ represents an exponential distribution. The constants (θ, τ) are defined as follows,

$$\theta = \frac{1-2p}{p(1-p)} \quad \text{and} \quad \tau = \sqrt{\frac{2}{p(1-p)}}.$$

The normal-exponential mixture representation of an AL distribution offers access to the properties of normal distribution, which can be exploited to construct a Gibbs sampler. Based on the above setting, answer the following questions.

- (a) Derive the likelihood function $f(y|\beta, w)$ and write the expression for the joint posterior distribution $\pi(\beta, w|y)$. Assume the parameter β follows a multivariate normal prior distribution i.e., $\beta \sim N(\beta_0, B_0)$.
- (b) Derive the conditional posterior distribution for β .
- (c) Derive the conditional posterior distribution for w .

Hint: A random variable z is said to follow a generalized inverse Gaussian (GIG) distribution, i.e., $z \sim GIG(\nu = \frac{1}{2}, a, b)$ if the density function is given by,

$$f(z) = \frac{(b/a)^{\frac{1}{2}}}{2K_\nu(ab)} z^{(\frac{1}{2}-1)} \exp \left[-\frac{1}{2}(a^2 z^{-1} + b^2 z) \right], \quad z > 0, -\infty < \nu < \infty, a, b, \geq 0,$$

and $K_\nu(\cdot)$ is a modified Bessel function of the third kind.

Bayesian Econometrics (ECO545A), Semester II 2023-24

Final Exam (Open Book) – Total Points: 100

Instructor: M.A. Rahman

5:00 - 8:00 pm, April 28, 2024

Please read the instructions carefully

- Please use the IITK answer sheet only.
- Questions should be answered in order as they appear in the exam. Every new question should begin in a new page.
- Full points will be awarded only if the answers are correct and to the point.

1. (4+8+8 = 20 points). Explain the following.

- (a) Define a Markov chain and Markov chain simulation.
- (b) What is Gibbs Sampling and when it used? Explain Gibbs sampling in the context of linear regression, $y = X\beta + \epsilon$, where $\epsilon \sim N(0, \sigma^2 I_n)$ and priors are $\beta \sim N(\beta_0, B_0)$ and $\sigma^2 \sim IG(\alpha_0/2, \delta_0/2)$.
- (c) What is Metropolis-Hastings (MH) algorithm and when it is used? Explain the difference between random-walk MH algorithm and tailored MH algorithm.

2. (5+5+10 = 20 points). Consider Question 1 from Homework 4 where the objective was to fit a binary quantile regression to 'public opinion that a course taken online provided an equal educational value compared with a course taken in person in a classroom' based on individual, socioeconomic, and demographic variables. The results from fitting a binary quantile regression at the 10th, 25th, 50th, 75th, and 90th quantiles are presented in Table 1 along with the covariate effects in Table 2.

- (a) Comment on the effect of variable 'online course' on $\Pr(y) = 1$ at a chosen quantile.
- (b) How will you interpret the coefficient for 'Post-Bachelors' on $\Pr(y) = 1$ at a chosen quantile?
- (c) Explain how will you compute the average covariate effect for 'online course' at the 10th quantile.

3. (5+5+5+5+5+5 = 30 points). Consider the logit model,

$$z_i = x_i' \beta + \epsilon_i, \quad \forall i = 1, \dots, n,$$

$$y_i = \begin{cases} 1 & \text{if } z_i > 0, \\ 0 & \text{otherwise,} \end{cases}$$

where $\epsilon_i \stackrel{iid}{\sim} \mathcal{L}(0, \pi^2/3)$ and $\mathcal{L}$ denotes the standard logistic distribution. The components within the parenthesis are the mean and variance. Based on the model, answer the following.

(a) Find $P(y_i = 1 | x_i' \beta)$.

(b) Find the log of the odds ratio i.e., $\ln[P_i/(1 - P_i)]$, where $P_i = P(y_i = 1 | x_i' \beta)$. Give your comments.

(c) Find the full likelihood for the logit model.

(d) Find the full posterior distribution of the logit model assuming a normal prior distribution i.e., $\beta \sim N(\beta_0, B_0)$. Can you construct a Gibbs sampler from the full posterior distribution?

(e) Find the expression for the augmented joint posterior i.e., $\pi(\beta, z | y)$.

latent (f) Does including the latent data z in the model helps in the construction of the conditional posteriors for β and z ? Explain. How will you estimate the model using MCMC?

4. (5+5+5+5+5+5 = 30 points). Consider the binary probit longitudinal data (BPLD) model expressed as a function of covariates with common effects and individual-specific effects. The BPLD model can be conveniently expressed in the latent variable formulation as follows,

$$z_{it} = x_{it}' \beta + s_{it}' \alpha_i + \epsilon_{it}, \quad \forall i = 1, \dots, n, \quad t = 1, \dots, T_i,$$

$$y_{it} = \begin{cases} 1 & \text{if } z_{it} > 0, \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where the latent variable z_{it} denotes the value of z at the t -th time period for the i -th individual, x_{it}' is a $1 \times k$ vector of explanatory variables, β is $k \times 1$ vector of common parameters, s_{it}' is a $1 \times l$ vector of variables that have individual-specific effects, α_i is an $l \times 1$ vector of individual-specific parameters, and ϵ_{it} is the error term assumed to be independently and identically distributed (iid) as $N(0, 1)$. Note that s_{it} may contain a constant for intercept heterogeneity, as well as other covariates (which are often a subset of those in x_{it}) to account for slope heterogeneity of those variables. The variable z_{it} is unobserved and represents the latent utility associated with the observed binary choice y_{it} which is related to z_{it} according to the second line of equation (1).

The stacked version of the model is often helpful for deriving the posterior densities of the regression coefficients and speeding up the computation. So, we define $z_i = (z_{i1}, \dots, z_{iT_i})'$, $X_i = (x_{i1}, \dots, x_{iT_i})'$, $S_i = (s_{i1}, \dots, s_{iT_i})'$, and $u_i = (u_{i1}, \dots, u_{iT_i})'$. The resulting hierarchical

model can be written as,

$$\begin{aligned} z_i &= X_i \beta + S_i \alpha_i + \varepsilon_i, \\ y_{it} &= \begin{cases} 1 & \text{if } z_{it} > 0, \\ 0 & \text{otherwise,} \end{cases} \\ \varepsilon_{it} &\sim N(0, 1), \quad \beta \sim N_k(\beta_0, B_0), \quad \alpha_i | \varphi^2 \sim N_l(0, \varphi^2 I_l), \quad \varphi^2 \sim IG(c_1/2, d_1/2), \end{aligned} \quad (2)$$

where the last three components in the last row represents the prior distributions with N and IG denoting the normal and inverse-gamma distributions, respectively. We use a simple independent normal prior for α_i , but it can be easily generalized to include a non-zero mean and a covariance matrix. For our purpose, let $T_i = T$ for all $i = 1, \dots, n$. Based on the above setting, answer the following.

- (a) Find $E(z_i)$ and $Var(z_i)$ marginal of α_i .
- (b) Write the augmented posterior distribution of the BPLD model.
- (c) Derive the conditional posterior of β marginal of α_i .
- (d) Derive the conditional posterior distribution of α_i .
- (e) Derive the conditional posterior for φ^2 .
- (f) Derive the conditional posterior distribution for z_i marginal of α_i . How will you sample the values of z_i . Explain.